Humanoid Robot Model for Space Survey

Chinnachart Pianpholdeesakul

Abstract

This research is a study to prototypes of d robot for light armored reconnaissance missions. In response to the country's security policy, the Thai military's policy on counterterrorism and the threat of terrorism is now in place. Military officers went on missions and lost their lives. In these missions, it is necessary to start researching and researching robots instead of humans in order to replace the lost or inadequate power. In the robot prototype, this research is primarily used for reconnaissance. And prevent the threats on the mission. The form of operation is a remote control within the location. The wireless signal is used to respond to the remote system as well.

Keywords : Humanoid Robot Model, anti- terrorism, Remote Weapon System and Wireless Network

1. Introduction

At present, terrorism is another formidable threat. Which the form of terrorism that occurs can be considered an off-war forms. Due to the inability to determine the attack position. Therefore must be prevented and monitored detect and observe. But in surveillance, it is necessary to use modern technology to cope with the mission. And in order to comply with the national security policy of research and development of appropriate technology for cop. That is the unmanned master robot for lightly armed reconnaissance missions. The goal of this research is to create the first robot prototype used to perform reconnaissance missions in various areas. And can prevent and also suppress threats to security. And can be applied to patrol missions along the border and the dense forest area is difficult to explore. Therefore, the attack pattern will change according to each terrain used in the mission. In addition, in conducting this research, studying to build the next prototype robot. The important part of this research is to design and simulate various work results. In accordance with the mission of the country. As well as to find a suitable way for work to respond to national defense technology policies and security.

2. Methodology

2.1 Research process

The research approach is divided into various parts of the robot. To guide the development of control systems and mechanisms. According to the structure of the robot

2.1.1. Head design mechanism

Beginning from the design of various angles and directions along the 6-axis independently (6 degree of freedom)

Joint		Range [°]
Eyes	pan	-45 to +45
	tilt	-40 to +40
Neck	pan	-55 to +55
	tilt	-50 to +40
	swing	-40 to +40

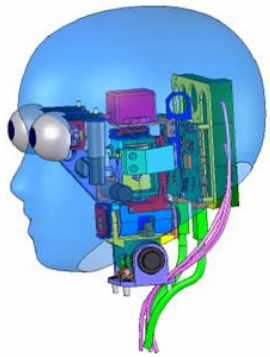

Figure 2.1 Virtual head and electric system

2.1.2. Algorithm of the robot head

1) Instead of the position image and vision

From $\widetilde{M} = [X \quad Y \quad Z \quad 1]^T$ and $\widetilde{m} = [X \quad Y \quad 1]^T$ in the derivative system, instead of the castle system M when M is $[X \quad Y \quad Z \quad 1]^T$ along the 3D axis and m is $[X \quad Y \quad 1]^T$and along the 2D axis is

$$s\widetilde{m} = A[Rt]\widetilde{M} \tag{2.1}$$

where S instead of the image, R and t. Instead of the position of the object. So, rotate around the degree axis of the value A is shown in the matrix form,

$$A = \begin{bmatrix} \alpha & \gamma & x_0 \\ 0 & \beta & y_0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2.2}$$

when α and β represents the value in the meter scale and γ instead of rotating along the 2-axis direction, the degree is (x_0, y_0).

2) The position of the eyes and retina

From the joint axis of the eye or the lens that looks at the image in 3 dimensions from 3D calculations based on the axis D_x , D_y is

$$D_x = \frac{\alpha_f X + \gamma_f Y}{Z} \tag{2.3}$$

$$D_y = \frac{\beta_f Y}{Z} \tag{2.4}$$

where

$$X = \frac{D_x - \gamma_f D_y / \beta_f}{\alpha_f} Z \tag{2.5}$$

$$Y = \frac{D_y}{\beta_f} \tag{2.6}$$

Instead of in the form matrix

$$s\begin{bmatrix}\hat{x}_p + d_x \\ \hat{y}_p + d_y \\ 1\end{bmatrix} = A_p\begin{bmatrix}(r_{11}(D_x - \beta_f D_y/\beta_f)/\alpha_f + r_{12}D_y/\beta_f + r_{13})Z - r_1 \cdot t \\ (r_{21}(D_x - \gamma_f D_y/\beta_f)/\alpha_f + r_{22}D_y/\beta_f + r_{23})Z - r_2 \cdot t \\ (r_{31}(D_x - \gamma_f D_y/\beta_f)/\alpha_f + r_{32}D_y/\beta_f + r_{33})Z - r_3 \cdot t\end{bmatrix}$$

2.1.3. Algorithm of the arm and body parts

The design in this robot arm is the positioning and direction according to Newton's law of 2-axes, with mass and force acting in the direction that,

$$F_1^0 = m_1 a_1^0,\ F_2^0 = m_2 a_2^0 \tag{2.7}$$

And from Euler's differential difference used to determine the dynamic of the torque point at the center of gravity and angular momentum, including the angular acceleration,

$$T_1^0 = I_1 \alpha_1^0 + \omega_1^0 \times (I_1 \omega_1^0),$$
$$T_2^0 = I_2 \alpha_2^0 + \omega_2^0 \times (I_2 \omega_2^0) \tag{2.8}$$

2.1.4. Algorithms of leg and motion systems

In a dynamic phase-based motion system with a pattern

$$\dot{\emptyset}_c = \omega_c + K_c \sin(\emptyset_r - \emptyset_c)$$
$$\dot{\emptyset}_r = \omega_r + K_r \sin(\emptyset_c - \emptyset_r) \tag{2.9}$$

Where $\omega_c > 0$ and $\omega_r > 0$, instead of the natural frequency of the robot's control and mechanism Which this model can find the balance of the system,

$$y = \frac{y_{foot}^t F_z^t + y_{foot}^r F_z^r}{F_z^t + F_z^r} \tag{2.10}$$

when F_z^t and F_z^r instead of the left and right reaction forces respectively.

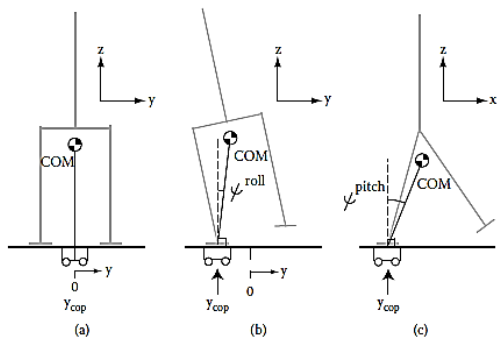

Fig. 2.2 Free degrees of freedom for both legs

The equation form and design model in the hip and knee joints are

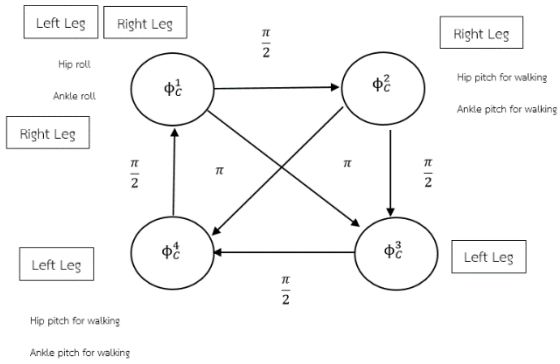

Fig 2.3. Connection points for each set of both leg joints

3. Connection points for each set of both leg joints

3.1 Visibility to depth in 3 dimensions

Research problems start with vision. By using the 3D image processing method based on the axial view, this image will set the setting to be a 3-axis reference axis based on the reference pattern $p_i = (a_i, b_i, c_i)^t, i = 1, \dots, n, n \geq 3$. Causes the image view based on the core to be considered $q_i = (a_i', b_i', c_i')^t$ By proving it in the form of an equation

$$q_i = Lp_i + w \tag{3.1}$$

Where

$$L = \begin{pmatrix} l_1^t \\ l_2^t \\ l_3^t \end{pmatrix} \epsilon\, SO(3) \text{ and } w = \begin{pmatrix} w_a \\ w_b \\ w_c \end{pmatrix} \epsilon\, R^3 \tag{3.2}$$

Instead of the rotation matrix and the use of translation vector, respectively, from the reference point that creates the image (image point) when $M_i = (o_i, k_i, 1)^t$ Perpendicular to the reference axis, representing

$$o_i = \frac{l_1^t p_i + w_x}{l_3^t p_i + w_z} \tag{3.3}$$
$$k_i = \frac{l_2^t p_i + w_y}{l_3^t p_i + w_z} \tag{3.4}$$

or

$$k_i = \frac{1}{l_3^t p_i + w_z}(Lo_i + w) \tag{3.5}$$

Which using the quality of the Colin Quality equation In finding the image of the core, it will be as shown in the figure. (3.1)

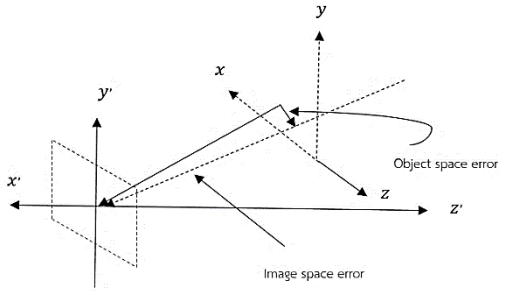

Fig 3.1 The picture shows the area of the object and the angle of the image that occurred.

Which the direction of the direction that occurs from the perpendicular direction along this axis is the part that supports the direction of various directions,

$$Lo_i + w = k_i(Lo_i + w) \tag{3.6}$$

Where

$$W_i = \frac{w_i w_i^t}{w_i^t w_i} \tag{3.7}$$

When compared with Line-of-sight Of the perpendicular light wave Which is the scene for receiving images Compared to W_i according to this feature is

$$\|x\| \geq \|W_i x\|, \quad x \epsilon R^3 \tag{3.7a}$$

$$w_i^t = w_i \tag{3.7b}$$

$$w_i^2 = w_i w_i^t = w_i \tag{3.7c}$$

From these axes, you can find the vertical distance of the image that is angled to the vertical axis of the axis as the equation (3.5) respectively.

3.2 Problems with the structure of the arms and hands in 6-axis degrees of freedom

Based on research problems of the past topics, it was found that the relationship of the vision system and the pilot conditions in the robot armed mechanisms with response from various parts, namely, dynamic position, degree of movement, speed Including directions for controlling the structure of the robot. When the relative elements that allows the robot arm to move in degrees of freedom by using inverse kinetics theory is the main conditions for controlling and designing various structures. Of the robot arm. The main goal are to be able to simulate the robot's arm mechanism and prove the results as shown in Figure 3.2.

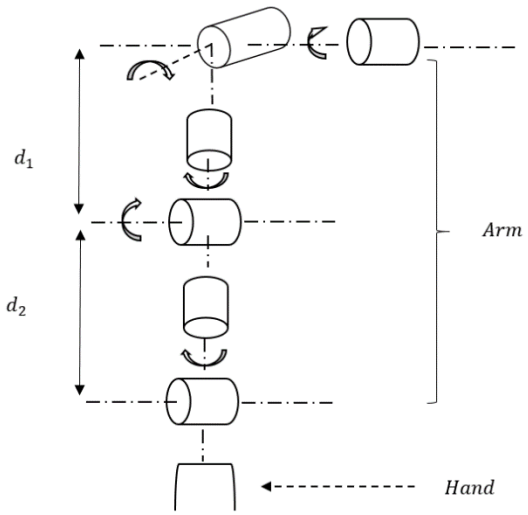

Fig 3.2 Independent dynamic structure design

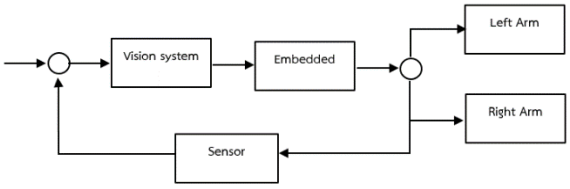

Fig 3.3 Independent dynamic structure design

With the 4 independent axes along the rotation When each corner is θ_i and various sliding axes are d_i when the distance is a_i and the angle between the axes z and α_i is

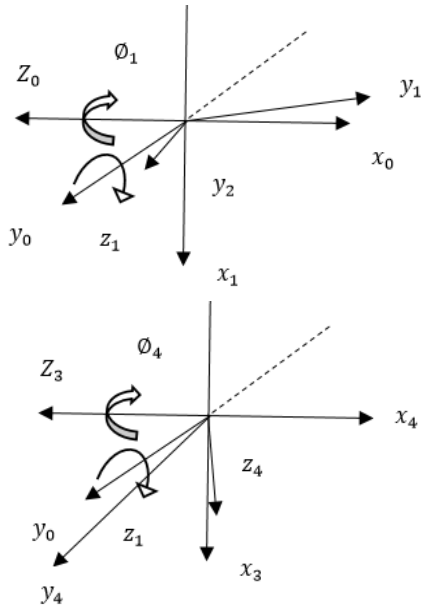

Fig 3.4 Shows the axis direction of the arm 180°

As shown in the form of linear homogeneous transformations when

$F_i\ (i = 1,2,3,\dots,6)$ will be

$$F_i^{i-1} = \begin{bmatrix} a\emptyset_i & -a\beta_i s\emptyset_i & b\beta_i s\emptyset_i & \varepsilon_i c\emptyset_i \\ b\emptyset_i & a\beta_i c\emptyset_i & -b\beta_i c\emptyset_i & \varepsilon_i s\emptyset_i \\ 0 & s\beta_i & c\beta_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3.8)$$

where $b\emptyset_i = \sin\emptyset_i,\ b\beta_i = \sin\beta_i, a\emptyset_i = \cos\emptyset_i,$ $a\beta_i = \cos\beta_i$

$$F_1^0 = \begin{bmatrix} a_1 & 0 & -b_1 & 0 \\ b_1 & 0 & a_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$F_2^1 = \begin{bmatrix} a_2 & 0 & b_2 & 0 \\ b_2 & 0 & -a_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$F_3^2 = \begin{bmatrix} a_3 & 0 & -b_3 & 0 \\ b_3 & 0 & a_3 & 0 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$F_4^3 = \begin{bmatrix} a_4 & 0 & b_4 & 0 \\ b_4 & 0 & -a_4 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$F_5^4 = \begin{bmatrix} a_5 & 0 & -b_5 & 0 \\ b_5 & 0 & a_5 & 0 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$F_6^5 = \begin{bmatrix} a_6 & 0 & -b_6 & 0 \\ b_6 & 0 & a_6 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Therefore, proving the inverse kinemetric,

$$[F_1^0(\emptyset_1)]^{-1}F_6^0 = F_2^1F_3^2F_4^3F_5^4F_6^5 = \begin{bmatrix} m_x' & r_x' & s_x' & g_x' \\ m_y' & r_y' & s_y' & g_y' \\ m_z' & r_z' & s_z' & g_z' \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3.9)$$

From the matrix in dimensions (1,4), (2,4), (3,4) in (3.9). To find the solution of the equation that

$$\begin{aligned} a_1 \cdot g_x + b_1 \cdot g_y &= g_x' \\ g_z &= g_y' \\ -b_1 \cdot g_x + a_1 \cdot g_y &= g_z' \end{aligned} \quad (3.10)$$

where $\cos\beta_4 = \left[g_x^2 + g_y^2 + g_z^2 - d_1^2 - d_2^2\right]/2d_1d_2$ with conditions $\beta_4 \epsilon [-180^\circ \sim -70^\circ]$ will get

$$\beta_4 = Atan2\left(\sqrt{1 - cos^2\beta_4}, \cos\beta_4\right) \quad (3.11)$$

when $Atan2(\cdot)$ instead of all four quadrants in the function when considering the homogeneous rotation matrix and the rotation of the matrix $m \cdot m = r \cdot r = s \cdot s = 1$

$$m \cdot s = m \cdot r = r \cdot s = 0 \quad (3.12)$$

Reformat the matrix

$$F_6^0 \cdot \left[F_6^5(\beta_6)\right]^{-1} \cdot \left[F_5^4(\beta_5)\right]^{-1} = F_1^0F_2^1F_3^2F_4^3 \quad (3.13)$$

From the matrix in dimensions (1,4), (2,4), (3,4) on both sides, two equations in (3.7) equal is

$$\begin{aligned} d_2 \cdot (m_x \cdot b_6 - r_x \cdot a_6) + g_x &= d_1a_1b_2 \\ d_2 \cdot \left(m_y \cdot b_6 - r_y \cdot a_6\right) + g_y &= d_1b_1b_2 \\ d_2 \cdot (m_z \cdot b_6 - r_z \cdot a_6) + g_z &= d_1a_2 \end{aligned} \quad (3.14)$$

So, this $[m_x, m_y, m_z]$ instead of perpendicular projection (Orthogonal vector), $[s_x, s_y, s_z]$ instead of twisting the Approach vector), $[r_x, r_y, r_z]$ instead of Orientation vector and $[g_x, g_y, g_z]$ instead of the vector position of the end-effect point (end-effector). From various elements When solving the matrix problem, it is related to the connection to the arms, hands and shoulders. Can show how to control balance in the form of equations that

$$\begin{aligned} a_3 \cdot b_4 &= \left(a_1 \cdot a_2 \cdot m_x + b_1 \cdot a_2 \cdot m_y - b_2 \cdot m_z\right)b_6 \\ &\quad + (a_1 \cdot a_2 \cdot r_x + b_1 \cdot a_2 \cdot r_y \\ &\quad - b_2 \cdot r_z)a_6 \\ -b_3 \cdot b_4 &= \left(-b_1 \cdot m_x + a_1 \cdot m_y\right)b_6 \\ &\quad + \left(b_1 \cdot r_x + a_1 \cdot r_y\right)a_6 \\ -b_4 \cdot a_5 &= \left(-a_1 \cdot b_2 \cdot m_x + b_1 \cdot b_2 \cdot m_y + a_2 \cdot m_z\right)a_6 \\ &\quad + \left(-a_1 \cdot b_2 \cdot r_x + b_1 \cdot b_2 \cdot r_y \right. \\ &\quad \left. + a_2 \cdot r_z\right)b_6 \end{aligned}$$

$$b_4 \cdot a_5 = \left(-a_1 \cdot b_2 \cdot s_x + b_1 \cdot b_2 \cdot s_y + a_2 \cdot s_z\right) \quad (3.15)$$

Which will find the solution of the algorithm from the q value that

$$\begin{aligned} q_1 &= \left(a_1 \cdot a_2 \cdot m_x + b_1 \cdot a_2 \cdot m_y - b_2 \cdot m_z\right)b_6 \\ &\quad + \left(a_1 \cdot a_2 \cdot r_x + b_1 \cdot a_2 \cdot r_y - b_2 \cdot r_z\right) \cdot a_6 \\ q_2 &= (b_1 \cdot m_x + a_1 \cdot m_x)b_6 + (b_1 \cdot r_x - a_1 \\ &\quad \cdot r_y) \cdot a_6 \\ q_3 &= \left(a_1 \cdot b_2 \cdot m_x - b_1 \cdot b_2 \cdot m_y - a_2 \cdot m_z\right)a_6 \\ &\quad + \left(a_1 \cdot b_2 \cdot r_x \cdot r_y\right) \cdot a_6 + (a_1 \cdot b_2 \cdot r_x - b_1 \cdot b_2 \cdot r_y \\ &\quad - a_2 \cdot r_z)b_6 \end{aligned}$$

$q_4 = a_1 \cdot b_2 \cdot s_x + b_1 \cdot b_2 \cdot s_y + a_2 \cdot s_z$

when $\emptyset_4 \epsilon [-180^{\circ} \sim -70^{\circ}]$ can show the calculation algorithm in conclusion that

$$\emptyset_3 = Atan2[q_2 sign(b_4),\ q_1 sign(b_4)] \quad (3.16)$$
$$\emptyset_5 = Atan2[q_4 sign(b_4),\ q_3 sign(b_4)] \quad (3.17)$$

3.3 The robot's leg and hip dynamics system

Based on the problem, research in this topic will prove the dynamic balance of the robot's leg and hip mechanisms. By which the mechanical design model as shown in the figure 3.5

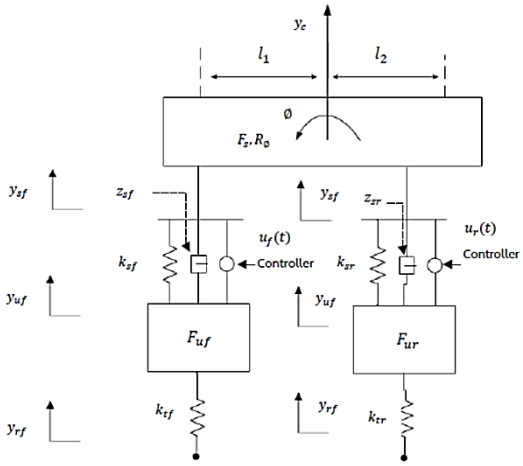

Fig 3.5 Showing the dynamic parts of the legs and hips

By the figure, instead of the value $y_{sf}(t)$. To be the core pitch of the robot body l_1 instead of the right mid-to-epic length l_2 instead of the mid-to-epic length left when compared to the center of mass (y_c). Give $\emptyset(t)$ instead of the raised angle or perpendicular (pitch angle) and give $y_c(t)$ instead of the displacement distance of the mass center. By which the mass of each robot leg represents F_{uf} and F_{ur} instead is the moment of inertia of the pitch, compared to the center of mass$(R_\emptyset)$ in this innovation, further studies have shown that lifting legs of robots that use actuators have a lot of pulling force, resulting in many test results obtained in the experiment. By which the mechanism of these loops Is defined as $F_{uf}(t)$ and $F_{ur}(t)$ respectively. When $y_{rf}(t)$ and $y_{rr}(t)$ instead of the length of the spring and torque that pulls the lower leg of the robot to be able to move according to various mechanisms. As defined While Newton's passive force was pressed down the control system that transmits feedback force is shaken back and forth, resulting in the total mass of vibrations. Therefore is a problem that needs to be solved in depth further and in the problems of this mechanical itself is entered by the mechanical results by giving $u_f(t)$ and $u_r(t)$ instead of the tension of actuators and various angles. Causing a swing 0° to 180° Which such angles represent $\emptyset(t)$ or a raised angle as a result of the spring rebounded force. Which causes small moment of inertia. Therefore must be corrected by finding the angle and force per moment of inertia that is done on the moment at the same time.

$$y_{sf}(t) = y_c(t) - l_1\emptyset(t) \quad (3.18)$$
$$y_{sr}(t) = y_c(t) - l_2\emptyset(t) \quad (3.19)$$

By finding that the force that actuator has overlapped is that the result is continuous

$$F_s\ddot{C}_c(t) + k_{sf}[C_{sf}(t) - C_{uf}(t)] + Z_{sf}[\dot{C}_{sf}(t) - \dot{C}_{uf}(t)] + k_{sr}[C_{sr}(t) - C_{ur}(t) + Z_{sr}]\dot{C}_{sr}(t) - \dot{C}_{ur}(t) = u_f(t) + u_r(t)$$
$$R_\emptyset\ddot{\emptyset}(t) - l_1k_{sf}[C_{sf}(t) - C_{uf}(t)] + l_1z_{sf}[\dot{C}_{sf}(t) - \dot{C}_{uf}(t)] + l_2k_{sr}[C_{sr}(t) - C_{ur}(t)] \quad +l_2z_{sr}[\dot{C}_{sr}(t) - \dot{C}_{ur}(t)] = -l_1u_f(t) + l_2u_r(t) \quad F_{uf}\ddot{C}_{uf}(t) - k_{sf}[C_{sf}(t) - C_{uf}(t)] + Z_{sf}[\dot{C}_{sf}(t) - \dot{C}_{uf}(t)] + k_{sf}[C_{uf}(t) - C_{sf}(t)] = -u_f(t) \quad (3.20)$$

From (3.18) to (3.20) causing the response of $\ddot{C}_{sf}(t)$ and $\ddot{C}_{sr}(t)$ will to,

$$\ddot{C}_{sf}(t) = \ddot{C}_c(t) - l_1\ddot{\emptyset}(t) = a_1\{u_f(t) - k_{sf}[C_{sf}(t) - C_{uf}(t)] - z_{sf}[\dot{C}_{sr}(t) - \dot{C}_{ur}(t)]\} + a_2\{u_r(t) - k_{sr}[C_{sr}(t) - C_{ur}(t)] - Z_{sr}[\dot{C}_{sr}(t) - \dot{C}_{ur}(t)]\} \quad (3.21)$$
$$\ddot{C}_{sr}(t) = \ddot{C}_c(t) - l_2\ddot{\emptyset}(t) = a_2\{u_f(t) - k_{sf}[C_{sf}(t) - C_{uf}(t)] - z_{sf}[\dot{C}_{sf}(t) - \dot{C}_{uf}(t)]\} + a_3\{u_r(t) - k_{sr}[C_{sr}(t) - C_{ur}(t)] - Z_{sr}[\dot{C}_{sr}(t) - \dot{C}_{ur}(t)]\} \quad (3.22)$$

where $a_1 = \frac{1}{F_s} + \frac{l_1^2}{R_\emptyset}, a_2 = \frac{1}{F_s} - \frac{l_1l_2}{R_\emptyset}, a_3 = \frac{1}{F_s} + \frac{l_2^2}{R_\emptyset}$

which we can find the results of the state-generated variables $x_1(t) = C_{sf}(t) - C_{uf}(t)$ when the sum of this force arises from the model of the lifting mechanism of the mechanics while $x_2(t) = C_{sr}(t) - C_{ur}(t)$ on the body of the robot or the answer to this is $x_3(t) = C_{uf}(t) - C_{rf}(t)$ which is vibrating from the legs of the robot that rebounded from the response of the 4th axis actuator $x_4(t) = C_{ur}(t) - C_{rr}(t)$ instead of the personal action of

the 5th axis robot $x_5(t) = \dot{C}_{sf}(t)$ or horizontal axis (yaw) of a robot that does not cause any shaking in the 6th axis $x_6(t) = \dot{C}_{sr}(t)$ instead of the line from the speed that collided from the robot to the vibration and the 7th axis $x_7(t) = \dot{C}_{uf}(t)$ be the speed axis to the 8th axis is $\dot{C}_{ur}(t)$. When disturbing conditions occur on the balance from the input voltage $w(t)$ of the system is $w(t) = \begin{bmatrix} \dot{z}_{rf}(t) \\ \dot{z}_{rr}(t) \end{bmatrix}$. And has a variable value that is a time state $x(t) = [x_1^T(t) \quad x_2^T(t) \quad x_3^T(t \quad x_4^T(t) \quad x_5^T(t)$ $x_6^T(t) \quad x_7^T(t) \quad x_8^T(t)]$ and $u(t) = \begin{bmatrix} u_f(t) \\ u_r(t) \end{bmatrix}$.

Where we calculate the balance of dynamics from friction in (3.20) and (3.21) instead of in the form of state-space equations,

$$\dot{x}(t) = Ax(t) + Bu(t) + B_1 w(t) \tag{3.23}$$

Where

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ -a_1 k_{sf} & -a_2 k_{sf} & 0 & 0 & -a_1 z_{sr} & -a_1 z_{sr} & a_1 z_{sf} & a_2 z_{sr} \\ -a_2 k_{sf} & -a_3 k_{sr} & 0 & 0 & -a_2 z_{sf} & -a_3 z_{sr} & a_2 z_{sf} & a_3 z_{sr} \\ \frac{k_{sf}}{F_{uf}} & 0 & \frac{k_{sf}}{F_{uf}} & 0 & \frac{z_{sf}}{F_{uf}} & 0 & -\frac{z_{sf}}{F_{uf}} & 0 \\ 0 & \frac{k_{sr}}{F_{ur}} & 0 & -\frac{k_{tr}}{F_{ur}} & 0 & \frac{z_{sr}}{F_{ur}} & 0 & -\frac{z_{sr}}{F_{ur}} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & a_1 & a_2 & -\frac{1}{F_{uf}} & 0 \\ 0 & 0 & 0 & 0 & a_2 & a_3 & 0 & -\frac{1}{F_{ur}} \end{bmatrix}^T$$

$$B_1 = \begin{bmatrix} 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \end{bmatrix}^T$$

Therefore, the active force that is designed will be performed in all directions in 3-axes, free degrees. This feature is caused by the force that actuators perform and pull out all the reflections. By which the pullback forced is stacked in the amount of speed and acceleration according to the output vector direction, explaining that $C_1(t) = \begin{bmatrix} \ddot{C}_c(t) \\ \ddot{\emptyset}_c(t) \end{bmatrix}$. From the design of this controller, it is the mechanism of the system that was developed and used to describe mechanical or mechanical mechanisms. With the force that is supported by this torque and spring has a constant balance under the second force on $|C_{sf}(t) - C_{uf}(t)| \leq C_{f_{max}}$,

$$|C_{sr}(t) - C_{ur}(t)| \leq C_{r_{max}} \tag{3.24}$$

Where $C_{f_{max}}$ and $C_{r_{max}}$ Represent the maximum and maximum acceleration from the shear system, respectively However, this result also corresponds to the mechanical balance of the robot that generates the response of this force and moment in the form of a load.

$$\left|k_{sf}\left(C_{uf}(t) - C_{rf}(t)\right)\right| \leq \xi_f \,,$$
$$\left|k_{sr}\left(C_{ur}(t) - C_{rr}(t)\right)\right| \leq \xi_r \tag{3.25}$$

when ξ_f and ξ_r be the static region of the load in the system variable Which we can calculate and solve the problem with the form of an algorithm that,

$$\xi_f(l_1 + l_2) = F_s g l_1 + F_{ur} g(l_1 + l_2) \tag{3.26}$$
$$\xi_f + \xi_r = (F_s + F_{uf} + F_{ur})g \tag{3.27}$$

In addition to the conditions in (3.24) and (3.25), we also found that we select the optimal value of the output from the internal forces that can be described as new.

$$\dot{x}(t) = Ax(t) + Bu(t) + B_1 w(t)$$
$$z_1(t) = C_1 x(t) + D_1 u(t)$$
$$z_2(t) = C_2 x(t) \tag{3.28}$$

where A, B and B_1 pulling the value of (3.23) will be

$$C_1 = \begin{bmatrix} \frac{k_{sf}}{F_s} & -\frac{k_{sr}}{F_s} & 0 & 0 & -\frac{z_{sf}}{F_s} & -\frac{z_{sr}}{F_s} & \frac{z_{sf}}{F_s} & \frac{z_{sr}}{F_s} \\ \frac{l_1 k_{sf}}{R_\phi} & -\frac{l_2 k_{sr}}{R_\phi} & 0 & 0 & \frac{l_1 z_{sf}}{R_\phi} & -\frac{l_2 z_{sr}}{R_\phi} & -\frac{l_1 z_{sf}}{R_\phi} & \frac{l_2 z_{sr}}{R_\phi} \end{bmatrix}$$

$$C_2 = \begin{bmatrix} \frac{1}{C_{f_{max}}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{C_{r_{max}}} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{k_{sf}}{\xi_f} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{k_{sr}}{\xi_r} & 0 & 0 & 0 & 0 \end{bmatrix} \tag{3.29}$$

$$D_1 = \begin{bmatrix} \frac{1}{F_s} & \frac{1}{F_s} \\ -\frac{l_1}{R_\phi} & \frac{l_2}{F_\phi} \end{bmatrix}$$

In the actuator force increased from this case study, causing the system to be interrupted continuously in the device and the controller that responds to this feedback will transform the logic motion of the signal into

$$\gamma(t) = k\varepsilon(t) \tag{3.30}$$

where k instead of designing a control variable that causes conditions under the variable, the ideal value of this balance are

$$\Delta K(t) = z\rho(t)D \tag{3.31}$$

when Z and D be a variable that is known and is a matrix of dimensions with dimensions $\rho(t)$ instead of the matrix of unknown functions, $\rho^T(t)\rho(t) \leq J$.

Therefore, the closed-loop system of state-space equations in the form of equations (3.30) instead of the new state equation is that

$$\begin{aligned} \dot{x}(t) &= Ax(t) + B_1 w(t) \\ z_1(t) &= \bar{C}_1 x(t) \\ z_2(t) &= C_2 x(t) \end{aligned} \tag{3.28}$$

So $\bar{A} = A + B\big(K + \Delta K(t)\big)$, $\bar{C}_1 = C_1 + D_1\big(k + \Delta K(t)\big)$. That means the algorithm is $w \epsilon L_2[0,\infty)$ without assumptions $\|w\|_2^2 \leq w_{max} < \infty$ When the goal is to design independent variables that can control variables with conditions that,

1) The balance will enter the stability point when the variable is right. Control is closed loops.

2) Under the condition that the center condition that this closed system becomes $\|z_1\|_2 < \gamma \|w\|_2$ Where in non-zero conditions the system will be $w \epsilon L_2[0,\infty)$ where $\gamma > 0$ instead of a scalar matrix.

3) The system output variable supported by the algorithm verification is that,

$$\left|\{z_2(t)\}_q\right| \leq 1, \;\; q = 1,2,3,4 \tag{3.29}$$

4. Problem solving in research

4.1 Guidelines for using algorithms on finding 3D images

In this research, we developed an algorithm to solve the problem of the image distance that occurred from the perspective on 3D using the capture method of the target or the object-space. That,

$$e_i = \big(I - \widehat{W}_i\big)(Ll_i + w) \tag{4.1}$$

where $\widehat{W}_i$ instead of the compensation from the path of the microwave source that is acted perpendicular to the linear matrix,

$$\widehat{W}_i = \frac{\widehat{W}_i \widehat{W}_i^t}{\widehat{W}_i^t \widehat{W}_i} \tag{4.2}$$

Representing the value with the lowest value when combined of the error values that,

$$\mathcal{F}(W,w) = \sum_{i=1}^{n} \|f_i\|^2 = \sum_{i=1}^{n} \left\|\big(I - \widehat{W}_i\big)\big(W_{p_i} + w\big)\right\|^2 \tag{4.3}$$

When $\mathcal{F}$ instead of finding the minimum value (minimize) W and w representing the variable of the offset position on the point of the image or the focusing distance that occurs $\{W_i\}$ that dealt with perpendicular matrices $\{\widehat{W}_i\}$ Which can find the most suitable point with a close-loop calculation.

$$w(W) = \frac{1}{n}\big(I - \frac{1}{n}\sum_j \widehat{W}_j\big)^{-1} \sum_j \big(\widehat{W}_j - I\big) W_{p_j} \tag{4.4}$$

From eq. (4.4) when $I - \frac{1}{n}\sum_{i=1}^{n} \widehat{W}_i$ we use the method of defining finite, which is a method of finding 3D vectors when $a \epsilon R^3$ will be,

$$\begin{aligned} & a^t\left(I - \frac{1}{n}\sum_{i=1}^{n} \widehat{W}_i\right) a \\ &= \frac{1}{n}\sum_{i=1}^{n} \left(\|a\|^2 - a^t \widehat{W}_i a\right) \\ &= \frac{1}{n}\sum_{i=1}^{n} \left(\|a\|^2 - a^t \widehat{W}_i^t \widehat{W}_i a\right) \\ &= \frac{1}{n}\sum_{i=1}^{n} \left(\|a\|^2 - \left\|\widehat{W}_i a\right\|^2\right) > 0 \end{aligned} \tag{4.5}$$

where $\|a\|^2 - \left\|\widehat{W}_i a\right\|^2$ and convert linear equations of functions $(\mathcal{L})$ and find this result as

$$q_i(\mathcal{L}) \stackrel{\text{def}}{=} \widehat{W}_i(\mathcal{L}p_i + w(\mathcal{L})) \text{ และ } \bar{q}(\mathcal{L}) \stackrel{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n} q_i(\mathcal{L}) \tag{4.6}$$

From this equation (4.3) can be written as,

$$\mathcal{F}(\mathcal{L}) = \sum_{i=1}^{n} \left\|\mathcal{L}p_i + w(\mathcal{L}) - q_i(\mathcal{L})\right\|^2 \tag{4.7}$$

With this case proving $\mathcal{L}$ from the matrix equation diagonally between $\{p_i\}$ และ $\{q_i(\mathcal{L})\}$ will be

$$\begin{aligned} M(\mathcal{L}) &= \sum_{i=1}^{n} q_i'(\mathcal{L})(p_i')^t \;\; \text{where } p_i' = p_i - \bar{p}_i \\ q_i'(\mathcal{L}) &= q_i(\mathcal{L}) - \bar{q}(\mathcal{L}) \end{aligned} \tag{4.8}$$

However We can calculate the value $\mathcal{L}$. By first assuming the estimation k from $\mathcal{L}$ is $\mathcal{L}^{(k)}, w^{(k)} = w\big(\mathcal{L}^{(k)}\big)$ and $q_i(k) = \mathcal{L}^{(k)} p_i + w^{(k)}$ or this form $\mathcal{L}^{(k+1)}$ To be used to solve the problem.

$$\begin{aligned} \mathcal{L}^{(k+1)} &= argmin_R \sum_{i=1}^{n} \left\|\mathcal{L}p_i + w\widehat{W} q_i q_i^{(k)}\right\|^2 \\ &= argmax_R wr\left(\mathcal{L}^t M\big(\mathcal{L}^{(k)}\big)\right) \end{aligned} \tag{4.9}$$

when $\widehat{W}_i q_i^{(k)}$ Used to find the hypotonicity of the spot on the scene q_i Which uses a matrix-like linear transformation method to calculate,

$$w^{(k+1)} = \mathrm{w}\left(\mathcal{L}^{(k)}\right) \qquad (4.10)$$

And can conclude that the algorithm used in the design of 3D images using post-editing methods solves this problem $\mathcal{L}^*$ is

$$\mathcal{L}^* = argmin_L \sum_{i=1}^{n} \left\| \mathcal{L}p_i + w - \widehat{W}_i\left(\mathcal{L}^* p_i + w(\mathcal{L}^*)\right) \right\|^2 \quad (4.11)$$

From the algorithm in the equation (4.11) and can design the algorithm for solving the image distance as shown in Figure 4.1

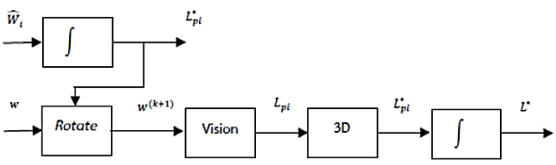

Fig.4.1 Algorithm for finding 3D images

4.2 Solving interference problems caused by in-depth images

In this way, we use the search for various effects of the incident point. Where the distance the image is distant from a distance and may encounter some interference caused by the gap between error. Which can find various compensation angles caused by various images signals Which we can bias and improve the optimal value using the algorithm is

$$\sum_{i=1}^{n} (\mathcal{L}p_i + w)^t (I - \widehat{W}_i) \Lambda_i^{-1} (I - \widehat{W}_i)(\mathcal{L}p_i + w) \qquad (4.12)$$

where $\Lambda_i$instead of the diagonal matrix used to cut the noise $\widehat{W}_i$ from the image signal that is affected by the point w_i .When the pattern of the linear diagonal matrix of the twisting distance is shown,

$$\begin{bmatrix} H(q) & 0 \\ M(q) & -w^t(q) \end{bmatrix} \begin{bmatrix} \dot{q} - q(F_p + \Delta F) \\ \mu - \Delta F \end{bmatrix} = 0$$

$$\begin{bmatrix} H(q) & -\mathcal{L}^t(q) \\ M(q) & 0 \end{bmatrix} \begin{bmatrix} \dot{q} - q(F) \\ \mu - \Delta F \end{bmatrix} = 0 \qquad (4.13)$$

5. Result

The results of experiments in each part of the robot have been processed. The results come out in various signal formats as follows.

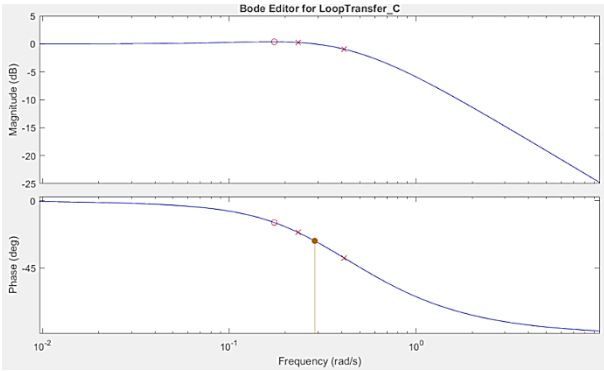

Fig. 5.1 Experimental results of frequency response signals.

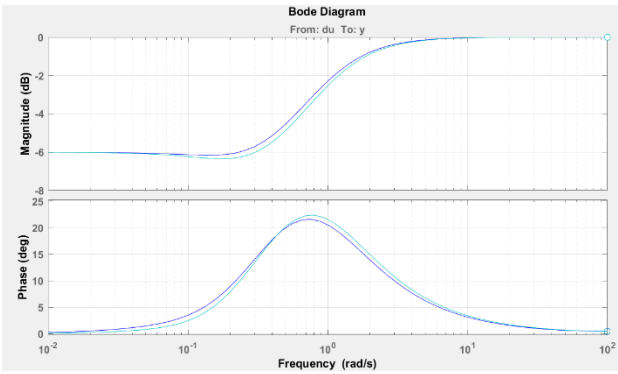

Fig. 5.2 Experimental results of time signals.

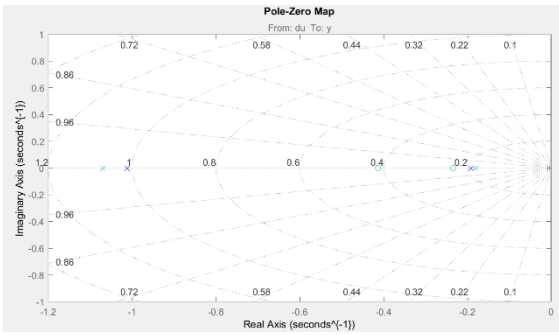

Fig. 5.3 Display the signal from the control system.

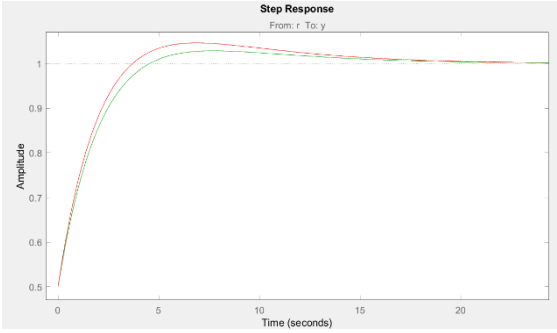

Fig. 5.4 Time response to the stability point

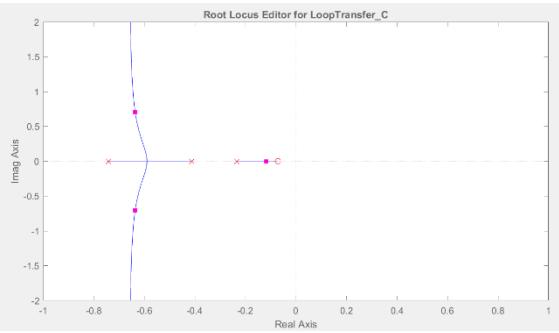

Fig. 5.5 Shows the root path of the system on the real axis

6. Conclusion

From the results of the experiment, we use the time response from the control system for the design of the variable of phases and magnitude that is expressed from the time signal. The results of the experiment show that dynamic calculations and results from this algorithm can prove the results of research and correspond to various theories of the way of mechatronics.

7. References

[1] Loi-Wah Sun, Frederick Van Meer, Yan Bailly, and Chung Kwong Yeung. Design and Development of a Da Vinci Surgical System Simulator. Proceedings of the 2007 IEEE International Conference on Mechatronics and Automation, Harbin, China. 2007:1050-1055.

[2] Jenny Dankelman. Surgical Robots and Other Training Tools in Minimally Invasive Surgery. 2004 IEEE International Conference on Systems, Man andCybernatics. 2004:2459-2464.

[3] Yun-Ju Lee, Jonathan Kim, Seong-Young Ko, Woo-Jung Lee, and Dong-Soo Kwon. Design of a Compact Laparoscopic Assistant Robot: KaLAR. ICCAS2003, Gyeongju, Korea.

[4] Russell H. Taylor and Dan Stoianavici. Medical Robotics in Computer Integrated Surgery. IEEE Transacctions on Robotics and Automation. 2003, 19(5):756-781

[5] Russell H. Taylor. A Perspective on Medical Robotics. Proceedings of the IEEE. 2006, 94(9):1652-1664.

[6] Yuan-Fang Wang, Darrin R. Uecker, Yu-Lun Wang. A New Framework for Vission-enabled and Robotically Assisted Minimally Invasive Surgery. Computerized Medical Imaging and Graphics. 1998:429-437.

[7] P.A.Finlay, M.H.Ornstein. Contrlling the Movement of a Surgical Laparoscope. IEEE in Medicie and Biology. 1995:289-291.